25901743 MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) –

MG4DSCPSY200 – FUNDAMENTALS OF COGNITIVE OPERATIONS

Duration: 1 Hour 30 Minutes (2024 ADMISSION ONWARDS) Maximum Marks: 50

Part A Multiple Choice Questions

Answer all questions. Each question carries 1 marks

1. Who adopted William Stern's formula for use with the translated and revised Binnet test at Stanford University? [U] / [CO1]
Lewis Terman – c

2. Behaviour that occurs as an automatic response to some stimulus are called ____[U] / [CO2]

Respondent Behaviour – c

3. One major reason eyewitnesses make errors is because of:[U] / [CO3]

Distortions in memory – a

4. Which factor improves memory retention?[U] / [CO3]

Overlearning – a

5. Why is a variable ratio schedule so effective at maintaining behavior?[U] / [CO2]

The unpredictability makes extinction difficult – b

6. Crystallized intelligence (G involves:[U] / [CO1]

Accumulated knowledge and verbal skills – c

7. The Bobo doll experiment was conducted by[K] / [CO2]

Albert Bandura – d

8. Explicit memory is also known as[K] / [CO3]

Declarative memory – b

9. The concept of the Language Acquisition Device[U] / [CO1]

Noam Chomsky – b

10. What refers to the fact that information entered into memory in one context or[U] / [CO3]

Context dependent memory – b

11 Gifted' individuals are typically defined as those falling in the top 2 percent of the population with the IQ of: [U] / [CO1]
130 or higher – c

12. What does the S in SQ3R stand for?[U] / [CO3]

Survey – a

13. According to Ebbinghaus, forgetting is:[U] / [CO3]

Rapid at first, then slows down– b

14. . Rules or step-by-step procedures that guarantees a solution to a specific type of [K] / [CO1]

Algorithms – b

15. Intelligence is best understood as the result of:[U] / [CO1]

Interaction between heredity and environment – c

[1x15 = 15]

Part B

Short Essay Questions.Answer any 5 questions.Each question carries 7 marks

16. Provide evidence for the existence of STM and LTM.[K] / [CO3]

Serial Position Effect: Research by Hermann Ebbinghaus and later studies show that when people are asked to recall a list of items, they tend to remember items at the **beginning (primacy effect)** and **end (recency effect)** better. The primacy effect reflects **LTM**, as early items are rehearsed and transferred to long-term storage. The recency effect reflects **STM**, as the last items are still present in short-term memory.

Word Length Effect: Short words are recalled more easily than long words, especially in immediate recall tasks. This suggests that STM has a **limited capacity based on time (about 2 seconds of rehearsal)**. Longer words take more time to rehearse and are therefore harder to maintain in STM, providing evidence for its limited duration and capacity.

In immediate recall (STM)**,** people tend to confuse **acoustically similar words** (e.g., cat, cap, can). This indicates that STM primarily encodes information acoustically (based on sound)**.** In delayed recall (LTM)**,** people are more likely to confuse **semantically similar words** (e.g., big, large, huge). This shows that LTM encodes information semantically (based on meaning)**.**

Note for valuation: Marks should also be awarded if students explain the differences between STM and LTM in terms of **duration and capacity**, and if any other **relevant, logical, and scientifically valid explanations** are provided, even if the answer is not strictly limited to the points mentioned.

17. What is the law of effect? Explain it in the context of trial and error learning.[U] / [CO2]

Thorndike's Trial-and-Error Theory of Learning. Trial and Error is a method of learning in which various responses are tentatively tried and some discarded until a solution is attained – stamping in of correct responses and stamping out of incorrect responses. 3 laws involved in trial and error learning: Law of readiness - A satisfying state of affairs results when an

individual is ready to learn and is allowed to do so , law of exercise- law which states that, in learning, the more frequently a stimulus and response are associated with each other, the more likely the particular response will follow the stimulus., law of effect - states that those behavioral responses that were most closely followed by a satisfying result were most likely to become established patterns and to occur again in response to the same stimulus.

18. Analyse how the neural network model explains memory.[An] / [CO3]

Neural network models suggest that it is the rich interconnectedness of our neural units that accounts for our ability to process information so quickly (see McClelland & Rumelhart,. These models also propose that information in memory is not located in a specific place within the brain; rather, it is represented by patterns of activation that spread over many processing units and by the strength of the activation across these various units. The analogy, offered by Lindsay and Reed is spider web. Think of neural networks as being like a spider web with millions of strands connecting various units. The tighter these strands, the stronger the connections among various units. Incoming information "pulls" on certain strands, thus activating other units, just as a fly that lands on a spider's web sends vibrations along the strands to other locations—including the center, where the spider is located.

19. Explain creativity ,its characteristics and stages[U] / [CO1]

The ability to produce work that is novel and appropriate. It should be original, unexpected, and useful one. It is a cultural, social and psychological event. The characteristcis include **Fluency** is defined as the total number of suggestions you are able to make. **Flexibility** is the number of times you shift from one class of possible uses to another. **Originality** refers to how novel or unusual your ideas are. **Divergent thinking** is the reverse, in which many possibilities are developed from one starting point.

The stages if creativity includes **Orientation.** As a first step, the person defines the problem and identifies its most important dimensions. **Preparation.** In the second stage, creative thinkers saturate themselves with as much information about the problem as possible. **Incubation.** Most major problems produce a period during which all attempted solutions will be futile. At this point,problem solving may proceed on a subconscious level:Although the problem seems to have been set aside, it is still “cooking” in the background. **Illumination.** The stage of incubation is often ended by a rapid insight or series of insights. These produce the “Aha!” experience, often depicted in cartoons as a light bulb appearing over the thinker’s head. **Verification.** The final step is to test and critically evaluate the solution obtained during the stage of illumination. If the solution proves faulty, the thinker reverts to the stage of incubation.

20. Define transfer of learning and different types of transfer.[U] / [CO2]

Transfer of learning refers to the influence of previous learning on the acquisition or performance of new learning. It explains how knowledge or skills learned in one context affect performance in another. Transfer can be of three main types. Positive transfer occurs when prior learning facilitates or improves new learning, such as when knowing how to ride a bicycle helps in learning to ride a motorcycle. Negative transfer occurs when earlier learning interferes with or hinders new learning, for example when habits of driving on one side of the road create difficulty when switching to the opposite side. Zero or neutral transfer occurs

when previous learning has no effect on new learning, such as when learning mathematics does not influence learning a musical instrument.

21. Define cognition. Explain the elements of thought. [U]/ [CO1]

Cognition refers broadly to mental processes or thinking.Thinking is an *internal representation* (mental expression) of a problem or situation. Thinking often involves all three units. **The elements are Images** are picture-like mental representations. mental representationsthat stand for objects or events and have a picturelike quality.**Concepts** are ideas that represent categories of objects or events. **Language** A language consists of symbols that convey meaning, plus rules for combining those symbols, that can be used to generate an infinite variety of messages.

22. Compare latent learning and insight learning with suitable examples. [U] / [CO2]

Latent learning is learning that remains hidden until its application becomes useful. Edward Tolman's latent learning experiment with rats demonstrated that learning can occur without immediate reinforcement and isn't always demonstrated behaviorally until a motivation (like food) is introduced.He used three groups of rats in a maze: one always rewarded, one never rewarded, and a third group rewarded only after 10 days; the third group showed rapid learning once food appeared, revealing they'd formed a mental "**cognitive map**" of the maze during their unrewarded explorations.
Wolfgang Kohler's Insight Learning Theory posits that problem-solving occurs through a sudden, cognitive restructuring of the environment, often called an "aha!" moment, rather than slow trial-and-error. Based on studies with chimpanzees, it emphasizes that learners perceive the whole situation and rearrange mental elements to reach a solution.
Köhler’s insight learning theory is based on his experiments with chimpanzees, which demonstrated that learning can occur through sudden understanding rather than gradual trial-and-error

[5×7= 35]